

Sparkup ERP

Predictive Attendance & Academic Lifecycle Platform

Track 3: ERP-Based Integrated Student Management System (Web Dev) | Event: WHYBE'26 Hackathon

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1. Problem Statement & Context

- **Spreadsheet & portal fragmentation**
 - Student records, marks, and rosters live across disconnected Excel sheets and legacy portals.
 - No single source of truth for admins, faculty, or students.
- **The hackathon / event attendance penalty**
 - Students at sanctioned off-campus events face automated exam debarment warnings.
 - Manual On-Duty (OD) reconciliation takes weeks – longer than the warning takes to fire.
- **No early-warning forecaster**
 - Attendance deficits surface only at exam registration – too late to recover.

2. Futuristic Innovations & Winning Features

- **Automated hackathon OD compensation engine**
 - 1-click OD claim filing for events (e.g., WHYBE'26).
 - Automatic attendance reconciliation the moment faculty verifies – no paperwork trail.
- **Predictive 75% attendance forecaster (exam eligibility radar)**
 - Live formula: $N = \max(0, \lceil 0.75 \times \text{Total} - \text{Attended} \rceil)$ – exact lectures needed to stay eligible.
 - Status tags update instantly: **Safe (>80%)**, **At-Risk (75–80%)**, **Debarred Danger (<75%)**.
- **Seamless 1-click role switcher**
 - Instant toggle between Student View and Faculty/Admin View – built for a fast, convincing judge walk-through.

3. Core MVP Modules (10-Hour Execution Scope)

- **Module 1 – Role-Based Auth & Profiles**
 - Supabase Auth with granular Admin / Faculty / Student permissions.
- **Module 2 – Centralized Student Directory (CRUD)**
 - Real-time rosters, enrollment status, branch, and semester course allocation.
- **Module 3 – Live Session-Wise Attendance Matrix**
 - Faculty roll-call interface, instant aggregate & subject-wise % calculation, OD reconciliation.
- **Module 4 – Analytics & Visualization Dashboard**
 - Radial 75% eligibility gauge and subject-wise comparison charts via Recharts.

4. Technology Stack & Architecture

- **Frontend:** Next.js 14 (App Router) with TypeScript for type-safe academic data models.
- **UI:** Tailwind CSS paired with shadcn/ui – responsive, dark-mode ready.
- **Database & security:** Supabase (PostgreSQL 17) with Row Level Security protecting grade and attendance records.
- **Visualization:** Recharts for attendance gauges and progress trends.
- **Deployment:** Vercel continuous deployment linked to GitHub (**staging** for testing, **main** for production).

5. Planned Live Demo Walkthrough for Judges

1. **Student login** – student opens the attendance dashboard for Engineering Mechanics.
2. **OD claim submission** – student files a 1-click compensation request for the WHYBE'26 Hackathon.
3. **Faculty one-click approval** – faculty switches roles, reviews the pending queue, approves.
4. **Live gauge recalculation** – dashboard updates to **76% (Eligible)** in real time, no page reload.